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End Semester Examination 2024

Name of the Course: B. Tech. Biotechnology

Semester: VII

Name of the Paper: Food and Nutrition Science

Paper Code: **TBT 705b**

Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Each sub question carries 10 marks

Q1	(20marks)	
(a)	Illustrate the pathways involved in the process of lactic acid fermentation.	CO2
(b)	Describe the processing and preservation technologies for milk and dairy products.	
(c)	Discuss the scope of microorganisms in food technology.	
Q2	(20 marks)	
(a)	What do you understand by the term irradiation? Explain all three-irradiation process of food preservation.	CO1
(b)	Define the distinction between primary and secondary fermentation in the creation of alcoholic beverages.	
(c)	Explain various ancient and traditional methods of food processing.	
Q3	(20 marks)	
(a)	Express the conceptual framework of catabolic suppression in the context of brewing.	CO3
(b)	Explain the terms High gravity brewing and Beta Glucans.	
(c)	Explain the effect of irradiation on food constituents.	
Q4	(20 marks)	
(a)	Construct the steps involved in beer production.	CO4
(b)	Explain the preservation of food in both low and high temperature.	
(c)	Mention the guidelines for the production, release, and movement of GMOs.	
Q5	(20 marks)	
(a)	Explain the terms (1) Probiotics and Prebiotics, (2) Single cell Protein.	CO5
(b)	Defend the significance of looking for organisms that have been metabolically injured in dairy-based products.	
(c)	Discuss various methods of detection of some food borne organisms.	

